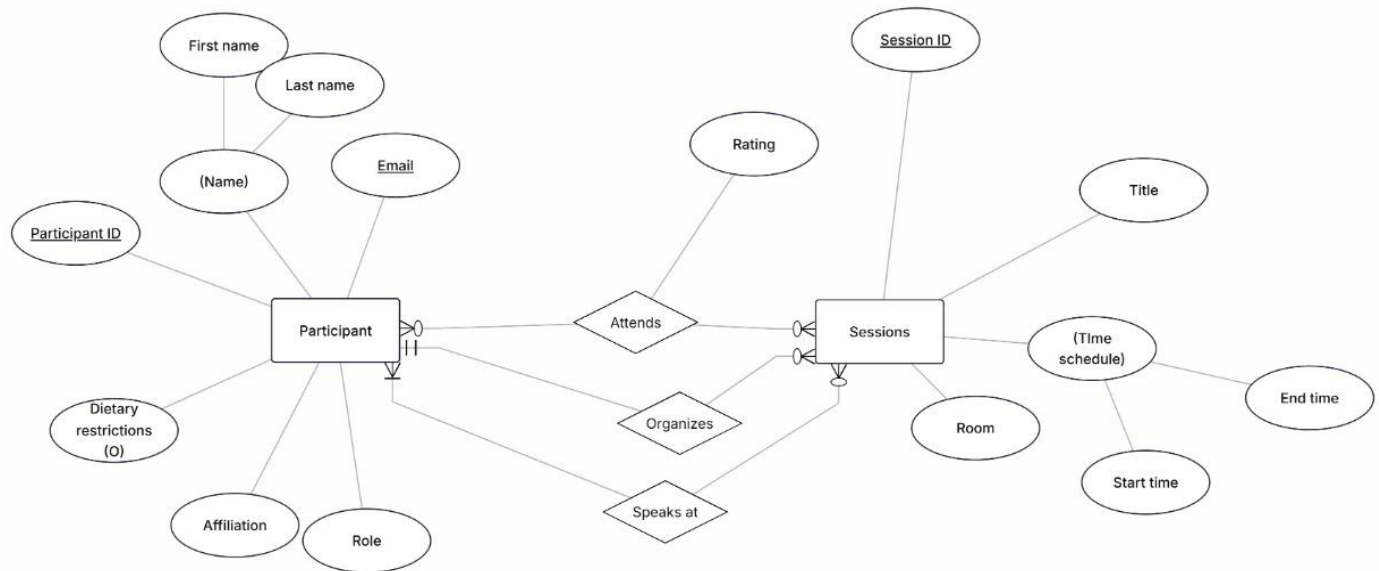


Task 1



Restrictions:

- Sessions run parallel to each other
- Three different types of participants - three different relationships
- Participants cannot attend two sessions that overlap in time
- Only regular participants register their attendance at each session
- Only regular participants rate the sessions

Assumptions:

1.Session:

- Each session has exactly one chair and happens in one specific room.
- Only one session can use a room at any time (no two sessions can happen in the same room at once).
- Sessions must have at least one speaker and can have more than one speaker
- Speakers can speak at one or many sessions; the sessions must not overlap

- A session can have zero regular participants and at most many
- Regular participants can attend multiple sessions, but the sessions can't overlap. A participant cannot be registered for two sessions that overlap in time

2.Participant:

- The 'organizes' relationship in the ERD represents the 'chair' role
- All chairs listed in the workshop program are registered as participants with the role 'organiser'
- Speakers and chairs are restricted from attending other sessions as regular participants or providing ratings
- We consider the dietary restriction text field as a single value

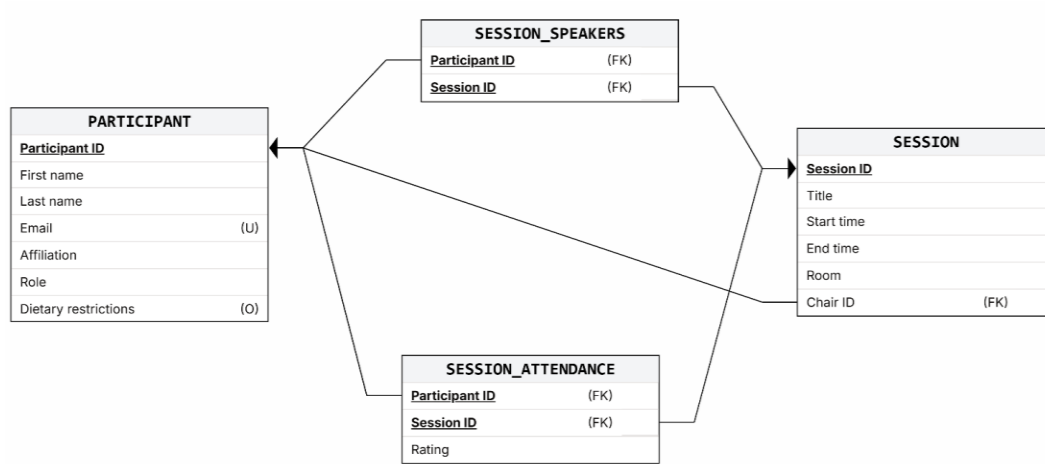
3.Attendance:

- A set of registered regular participants has no attendance records, allowing for the testing of the card registration system

4.Rating:

- The sample data includes attendance and ratings for all six sessions
- Every session has sample data for rating scores, but ratings are optional
- A few records for rating are NULL to represent participants whose rating has not been provided yet (forgetting or postponing to give the rating score)

Task 2



-- Task 3

PRAGMA foreign_keys = ON;

-- 1. Drop tables in reverse order (Children first to avoid FK errors)

DROP TABLE IF EXISTS Session_Attendance;

DROP TABLE IF EXISTS Session_Speakers;

DROP TABLE IF EXISTS Session;

DROP TABLE IF EXISTS Participant;

-- 2. Create Tables

```
CREATE TABLE Participant (  
    participant_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    first_name VARCHAR(255) NOT NULL,  
    last_name VARCHAR(255) NOT NULL,  
    email VARCHAR(255) NOT NULL UNIQUE,  
    affiliation VARCHAR(255),  
    role VARCHAR(50) NOT NULL CHECK (role IN ('organiser', 'speaker', 'regular_participant')),  
    dietary_restrictions TEXT  
);
```

```
CREATE TABLE Session (  
    session_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    title VARCHAR(255) NOT NULL,  
    start_time TIME NOT NULL,  
    end_time TIME NOT NULL,  
    room CHAR(6) NOT NULL,  
    chair_id INTEGER NOT NULL,  
    FOREIGN KEY (chair_id) REFERENCES Participant(participant_id)  
);
```

```
CREATE TABLE Session_Speakers (  
    session_id INTEGER,  
    participant_id INTEGER,  
    PRIMARY KEY (session_id, participant_id),  
    FOREIGN KEY (session_id) REFERENCES Session(session_id),  
    FOREIGN KEY (participant_id) REFERENCES Participant(participant_id)  
);
```

```
CREATE TABLE Session_Attendance (  
    participant_id INTEGER,  
    session_id INTEGER,  
    rating INTEGER CHECK (rating BETWEEN 1 AND 10),  
    PRIMARY KEY (participant_id, session_id),
```

```

FOREIGN KEY (participant_id) REFERENCES Participant(participant_id) ON DELETE
CASCADE,
FOREIGN KEY (session_id) REFERENCES Session(session_id) ON DELETE CASCADE
);

```

-- 3. Populate Participants (IDs 1-11 are Speakers/Orgs, 12-32 are Regulars)

```

INSERT INTO Participant (first_name, last_name, email, affiliation, role, dietary_restrictions)
VALUES
('Lina', 'Lian', 'lina.lian@uib.no', 'UiB', 'speaker', NULL),
('Maria', 'Chen', 'maria.chen@uio.no', 'UiO', 'speaker', NULL),
('Paul', 'Novak', 'paul.novak@mit.com', 'MIT', 'speaker', NULL),
('Omar', 'Haddad', 'omar.haddad@uio.no', 'UiO', 'speaker', NULL),
('Ivan', 'Petrov', 'ivan.petrov@uit.no', 'UiT', 'speaker', NULL),
('Marry', 'Jones', 'marry.jones@mit.com', 'MIT', 'speaker', NULL),
('James', 'Patel', 'james.patel@uio.no', 'UiO', 'speaker', NULL),
('Elena', 'Rossi', 'elena.rossi@ucla.com', 'UCLA', 'speaker', NULL),
('Jan', 'Kudlicka', 'jan.kudlicka@bi.no', 'BI', 'organiser', NULL),
('Magdalena', 'Ivanovska', 'magdalena.ivanovska@bi.no', 'BI', 'organiser', NULL),
('Wei-Ting', 'Yang', 'wei-ting.yang@bi.no', 'BI', 'organiser', NULL),
('Victoria', 'Lewis', 'Vicky_lewie@gmail.com', 'BI', 'regular_participant', 'Gluten Intolerance'),
('Jessica', 'Steveson', 'Jessica798@icloud.com', 'BI', 'regular_participant', NULL),
('Mark', 'Jacobs', 'Marcu.Jarcu00@facebook.org', 'BI', 'regular_participant', NULL),
('Stephanie', 'Melville', 'Brandy_melville_fan@bi.no', 'BI', 'regular_participant', 'Peanut Allergy'),
('Queen', 'Aslan', 'Aslan0123@gmail.org', 'BI', 'regular_participant', NULL),
('Lucy', 'Ishy', 'Lucy_flies010@foxnews.com', 'UiO', 'regular_participant', NULL),
('Urd', 'Trondheim', 'UrdT@gmail.com', 'UiO', 'regular_participant', NULL),
('Amelia', 'Chadewick', 'Ameleee@bi.no', 'BI', 'regular_participant', NULL),
('Grace', 'Lee', 'Graciousness@mit.no', 'MIT', 'regular_participant', 'Lactose Intolerance'),
('Faith', 'Tan', 'Faithfulness@mit.no', 'MIT', 'regular_participant', NULL),
('Syafiza', 'Chyena', 'jaSjaS@bi.no', 'BI', 'regular_participant', NULL),
('Torro', 'Freia', 'FreiaaaT@bi.no', 'BI', 'regular_participant', 'Vegan'),
('Svalbardian', 'Herpep', 'herpes@bi.no', 'BI', 'regular_participant', 'Vegetarian'),
('Joseph', 'Stepherson', 'JoeJoseph@bi.no', 'BI', 'regular_participant', 'Vegetarian'),
('Thorv', 'Thon', 'ThornEmo@bi.no', 'BI', 'regular_participant', 'Lactose intolerant'),
('Emin', 'Tragic', 'tragedyEm@bi.no', 'BI', 'regular_participant', 'Peanut Allergy'),
('Steve', 'Roberts', 'setevsnor@bi.no', 'BI', 'regular_participant', 'Peanut Allergy'),
('Peder', 'Diesmos', 'PedoDies@bi.no', 'BI', 'regular_participant', 'Lactose Intolerance'),
('Ben', 'Marcus', 'Ben@mit.no', 'MIT', 'regular_participant', NULL),
('Max', 'Syphillus', 'Max@gmail.com', 'BI', 'regular_participant', NULL),
('Henderson', 'Martial', 'Hendersi@gmail.com', 'BI', 'regular_participant', NULL);

```

-- 4. Insert Sessions (Jan=9, Magdalena=10, Wei-Ting=11)

```

INSERT INTO Session (title, start_time, end_time, room, chair_id) VALUES
('AI Ethics and Accountability', '09:00', '10:00', 'B2-010', 9),

```

('AI in Healthcare: Opportunities & Risks', '09:00', '10:00', 'B2-020', 10),
('AI, Privacy, and Surveillance in the Digital Age', '10:30', '11:30', 'B2-030', 11),
('AI and the Future of Work', '10:30', '11:30', 'C2-095', 10),
('AI Governance and Regulation', '13:00', '14:00', 'C2-055', 9),
('AI in Education and Learning Systems', '13:00', '14:00', 'C2-045', 11);

-- 5. Link Speakers

INSERT INTO Session_Speakers (session_id, participant_id) VALUES
(1,1), (2,2), (2,3), (3,4), (4,5), (4,6), (5,7), (6,8);

-- 6. Insert Attendances (Excluding IDs 30, 31, 32 for Task 5 testing)

INSERT INTO Session_Attendance (participant_id, session_id, rating) VALUES
(12, 1, 7), (17, 1, 4), (29, 1, 7), -- Session 1
(15, 2, 6), (19, 2, 8), (21, 2, 5), -- Session 2
(14, 3, 8), (21, 3, 7), (23, 3, 10), -- Session 3
(16, 4, 5), (24, 4, 8), (25, 4, 7), -- Session 4
(12, 5, 9), (18, 5, 6), (27, 5, NULL), -- Session 5 (NULL for testing task 6a)
(15, 6, 7), (21, 6, 5), (22, 6, NULL); -- Session 6 (NULL for testing task 6a)